

Global Energy Storage

Neil Beveridge, Ph.D. +852 2123 2648 neil.beveridge@bernsteinsg.com

Brian Ho, CFA +852 2123 2615 brian.ho@bernsteinsg.com

Katherine Li +852 2123 2694 katherine.li@bernsteinsg.com

Kelvin Yuan, Ph.D., CFA +852 2123 2612 kelvin.yuan@bernsteinsg.com

Battery Weekly 28 September

America

[CATL's New Batteries Target American Pickups. They Could Be Made In The U.S. - insideevs.com](#) CATL has developed “tall” battery cells for U.S. electric pickups and says unnamed American automakers have tested them. The company is considering licensing the technology for domestic production as a way to enter the market despite trade barriers. It has not disclosed the cells’ chemistry or specifications, named a customer, or announced a production timeline; U.S. manufacturing remains a possibility, not a confirmed plan.

[General Motors introduces recycled battery materials in customer vehicles- electrive.com](#) General Motors has put battery cells containing recycled cathode materials into customer vehicles, moving beyond prototype testing. In the pilot, Cirba Solutions processed 80 retired GM EV batteries into more than 12 tonnes of cathode active material, with all of its nickel, cobalt and manganese sourced from recycled material. GM–LG Energy Solution joint venture Ultium Cells made the new cells, which are now used in selected Cadillac, Chevrolet and GMC models. GM says they performed comparably to cells made with newly mined materials, though the project remains a pilot rather than a large-scale rollout.

[Onsemi aims to quadruple power density of EV inverters- electrive.com](#) Onsemi has introduced its Embedded Power Platform, which uses the silicon wafer as the package to integrate power devices and control components. The company says the architecture could deliver up to four times the power density and 15% lower power losses in EV traction inverters, potentially making them smaller and lighter. Subaru is evaluating the platform for future vehicles, but has not committed to production use; initial samples are expected in 2026.

[China's Eve Energy Surges After Unit Inks Major Deal to Supply Batteries to Fluence Energy- yicaiglobal.com](#) EVE Energy's subsidiary Hubei EVE Power has signed a five-year battery supply framework with US-based energy storage integrator Fluence covering up to 206 GWh from 2027 to 2031. The companies have committed to 16 GWh of deliveries in 2027, while the remaining 190 GWh is capacity reserved for 2028–2031, not a firm order; product specifications, pricing and delivery schedules will be set through subsequent purchase orders. The agreement could provide EVE with a substantial long-term channel into Fluence's global storage business, but its eventual revenue contribution depends on how much of the reserved volume converts into orders. For context, EVE shipped 71.05 GWh of energy storage batteries in 2025, making even the initial 16 GWh commitment material relative to last year's shipments. Its shares closed 6.9% higher at CNY51.56 on the announcement day.

Asia

[Korea Expands ESS Tender to 1,180 MW as Battery Makers Vie for 2 Trillion Won in Orders.- thelec.net](#) South Korea has opened a 1,180 MW centralized ESS tender—1,100 MW on the mainland and 80 MW on Jeju—more than double the 540 MW offered in last year's first round. Bidding closes on November 11, with winners expected that month and projects due by February 2029. The expanded tender could generate more than KRW2 trillion in orders, intensifying competition among Samsung SDI, SK On and LG Energy Solution; Samsung SDI led the first two rounds combined with 56% of awarded battery volume, followed by SK On at 25% and LGES at 19%. Jeju will also offer a 25-year contract option for systems capable of two charge-discharge cycles per day, potentially favoring longer-life batteries.

[Samsung SDI to Resume Synergy Cells Investment, Target 2028 Start.- thelec.net](#) Samsung SDI is reportedly preparing to resume investment in its Synergy Cells plant in Indiana, which became wholly owned after it acquired GM's stake. The company has confirmed that the plant will initially serve the U.S. energy storage market, while The Elec reports that Samsung SDI is considering one or two LFP production lines, each

potentially exceeding 10 GWh of annual capacity, with operations targeted for the second half of 2028. The report also estimates that Samsung SDI's 2027 capex could reach around KRW4 trillion and identifies Tesla as a possible end customer. Those line specifications, capex plans, customer discussions and start-up timing have not been confirmed by Samsung SDI.

[LG Energy Solution Begins Second Phase of Gwacheon R&D Campus Expansion.- thelec.net](#) LG Energy Solution has begun the second phase of its Gwacheon R&D campus expansion in South Korea, adding a research building as it broadens work beyond battery packs and battery management systems to include materials. The planned building has roughly 22,425 square metres of floor space, with approval for use targeted by end-2028 and research operations expected to begin in 2029. The expansion reflects continued investment in battery R&D despite a weaker industry backdrop, but it is a research facility rather than new cell-production capacity.

[EcoPro in Talks With CATL on Cathode Supply Deal.- thelec.net](#) EcoPro BM is reportedly in talks to supply CATL with around 20,000 tonnes of cathode material a year from its Debrecen plant in Hungary, with deliveries potentially starting in 2028. Using the report's assumption of 1,500 tonnes per GWh, that volume would support roughly 13 GWh of battery production. The proposed deal could give CATL a nearby European source of high-nickel cathodes and help EcoPro BM diversify beyond its existing customers. EcoPro BM also plans to increase the plant's annual capacity from 54,000 to 60,000 tonnes, but no CATL supply contract has been finalized; pricing and volume remain under discussion.

[Geely counters BYD with 2.25 MW charging system- electrive.com](#) Geely has unveiled a fifth-generation Smart Charging system with peak output of 2.25 MW, paired with its 12C-capable Shendun Golden Battery. The company says Lynk & Co 10 and Zeekr 001 vehicles charged from 10% to 70% in 4 minutes 30 seconds, and to 97% in 8 minutes 40 seconds, in its tests. An AI system adjusts charging power using predicted battery temperatures, while liquid cooling manages heat across the charger, cable and vehicle. Geely also claims its charging controls and pulse-restoration technology can extend battery cycle life by up to 20%, though it has not disclosed the test conditions or a rollout schedule for the new chargers.

[Hyundai and LGES launch battery storage pilot using repurposed EV batteries- electrive.com](#) Hyundai and Kia, together with LG Energy Solution and two other partners, have launched a 200-kWh battery-storage pilot at Wonik PNE's facility in Hwaseong, South Korea. Built from retired battery packs from Hyundai and Kia E-GMP electric vehicles, the system stores lower-cost electricity during off-peak hours and supplies it to DC fast chargers when demand is higher. The partners will test battery safety and performance, charging reliability and operating economics to assess whether repurposed EV batteries can support a viable charging-station business model; commercial-scale deployment has not yet been announced.

[Sunwoda to integrate 15C battery into megawatt charging ecosystem- electrive.com](#) Sunwoda is pairing its 15C-capable LFP battery with a planned megawatt-charging network, targeting 10,000 stations by the end of 2027. The company says the battery can charge from 10% to 70% in five minutes and to 97% in nine minutes at room temperature, or reach 97% in 15 minutes at -20°C. Its proposed ecosystem would combine fast chargers with on-site solar and battery storage and support vehicles from multiple brands. The 15C figure is a peak charging rate, not the rate sustained throughout a session; Sunwoda has not disclosed the network's locations or investment, or named a production vehicle using the battery.

[Solid-state batteries: Factorial leverages Japanese electrolyte expertise- electrive.com](#) Factorial Energy has signed a joint development agreement with Japan's Mitsui Kinzoku to help industrialize its Solstice all-solid-state battery platform. Mitsui will contribute expertise in sulfide-based solid electrolytes, while Factorial will lead cell design, manufacturing-process development and production validation. The partnership addresses a key materials and scale-up challenge, but the companies have not disclosed an investment amount, production capacity or commercialization timetable. Factorial's Solstice cells remain at the development stage, so the agreement should not be read as a mass-production announcement.

[China state media rebuts 'de-CATLization,' warns of battery supply chain price competition.-CNEVPOST](#) A commentary by CNII, a media outlet affiliated with China's Ministry of Industry and Information Technology, pushed back against the "de-CATLization" narrative, arguing that automakers adding battery suppliers or developing their own technology should not be framed as abandoning CATL. It warned that procurement focused too heavily on low prices could compromise battery quality and safety, and urged automakers to distinguish genuine in-house cell development from pack design, joint development or supplier manufacturing. Despite broader supplier diversification, CATL remained China's largest power-battery supplier, with a 41.45% installation share in August. This was a media commentary, not a new government policy.

[CBAK Energy Signs Agreement for RMB3 Billion, 12 GWh Sodium-Ion Battery Project in Nanjing.- solarbe.com](#) CBAK Energy has signed an investment agreement with the Nanjing Gaochun Economic Development Zone for a planned 12 GWh-per-year project producing large cylindrical sodium-ion battery cells and integrated battery systems. The company's Chinese announcement puts planned investment at approximately RMB3 billion and estimates annual sales of RMB4.8–5.0 billion once the project is fully built and operating. CBAK plans to begin advancing the project in the second half of 2027, subject to financing. Its proposed production lines could make either sodium-ion or lithium-ion cells, with the chemistry selected according to demand; customer testing of sodium-ion samples is still underway. The projected sales are

not contracted revenue and, according to CBAK's English release, use current lithium-ion cell prices as a benchmark.

Europe

[CATL's Hungary Plant Starts Trial Production of Battery Cells.- yicaiglobal.com](#) CATL began trial production on the first two battery-cell lines at its Debrecen plant in Hungary on September 22, after obtaining the permits required for trial operations. The plant is designed for an eventual 100 GWh of annual capacity and is expected to become CATL's largest manufacturing base outside China, but the current trial runs are for equipment and process validation—not commercial-scale cell production. The start follows regulatory scrutiny over worker nickel exposure: Hungarian authorities had suspended work in parts of the site in August, and CATL says the deficiencies identified during permitting have since been resolved. A start date for series cell production has not been announced.

[China's Yuneng Boosts Spain Battery-Materials Investment Nearly 70% as Capacity Grows.- yicaiglobal.com](#) Chinese cathode-materials maker Yuneng plans to raise investment in its Spanish project by nearly 70%, from CNY982 million to approximately CNY1.65 billion (USD253 million), while increasing planned annual capacity by 40%, from 50,000 to 70,000 tonnes. The company says the expansion reflects market demand and will support its international business; it plans to fund the additional investment through internal and other financing sources. Yuneng has not disclosed a revised start-up date or customer commitments for the enlarged facility.

[Cummins to manufacture LFP batteries for commercial vehicles in Spain- electrive.com](#) Cummins plans to manufacture LFP battery packs for electric buses and trucks at a new facility in Spain under its Accelera brand, strengthening its local supply to European commercial-vehicle customers. The Advanced LFP platform includes modular 102-kWh packs operating at up to 840 volts, alongside a smaller 51-kWh variant. Cummins has not disclosed the factory's location, investment, capacity or start-up date, and it remains unclear whether the site will make battery cells or assemble packs using externally supplied cells. European market launch has reportedly been targeted for 2028, but Cummins has not confirmed a timetable.

Others

[China's Zijin Longking to Invest USD232 Million in Solar-Storage Projects at South American Gold Mines.- yicaiglobal.com](#) Zijin Longking plans to invest approximately USD232 million in three solar-plus-storage projects at Zijin Mining-controlled gold mines in Suriname and Guyana, aiming to reduce reliance on costly diesel generation as mining operations expand. The largest, a USD150 million second-phase project at Suriname's Rosebel mine, will add 170 MWp of solar capacity and 120 MW/120 MWh of battery storage, with completion targeted for Q3 2027. Two projects at Guyana's Aurora mine will cost approximately USD55 million and USD26.6 million, respectively, and are targeted for Q4 2027. Together, the three projects would add 190 MW/300 MWh of battery storage; all electricity generated will be used at the mines. Investment amounts, timelines and expected output remain estimates, with overseas approvals still in progress.

EXHIBIT 1: **Key commodities price performance**

	Price 25-Sep	Ccy	Mkt cap \$bn	P/E 26E	EV /EBITDA 26E	%1D	%1W	%1M	%1Y	1Y px chart
Metal prices (US\$/tonne)										
LiCO (spot)	19,588					-1.5%	1%	-14%	90%	
LiOH (spot)	18,098					-1.6%	1%	-15%	75%	
LiCO (contract)	19,750					0.0%	0%	-1%	108%	
LiOH (contract)	18,250					0.0%	0%	-6%	102%	
Cobalt (spot)	39,209					0.0%	0%	-30%	13%	
Nickel (spot)	16,501					0.1%	2%	-3%	8%	
Component prices										
Cathode NMC811						0.0%	0%	-9%	23%	
Cathode LFP						0.0%	0%	-12%	46%	
Precursor NMC811						0.0%	0%	-6%	5%	
Precursor LFP						0.0%	0%	2%	52%	
Artificial Graphite						0.0%	0%	0%	6%	
Natural Graphite						0.0%	0%	0%	0%	
Electrolyte NMC						0.0%	0%	2%	38%	
Electrolyte LFP						0.0%	0%	3%	59%	
Separator						0.0%	0%	0%	28%	
Spot battery cell cost (US\$/kWh)										
LFP	59					0.0%	0%	-3%	14%	
NMC532	80					0.0%	0%	-6%	14%	
NMC622	76					0.0%	0%	-6%	9%	
NMC811	80					0.0%	0%	-5%	13%	
Spot battery pack cost (US\$/kWh)										
LFP	72					0.0%	0%	-2%	-2%	
NMC532	111					0.0%	0%	-4%	10%	
NMC622	109					0.0%	0%	-4%	6%	
NMC811	112					0.0%	0%	-4%	9%	

Source: Bloomberg, Baiinfo, Bernstein analysis and estimates (cell and pack cost)

EXHIBIT 2: Key companies price performance and valuation

	Price		Mkt cap	P/E	EV					
	25-Sep	Ccy	\$bn	26E	/EBITDA	%1D	%1W	%1M	%1Y	1Y px
					26E					chart
Cell makers										
LGES	351,500	KRW	60.4	n.a.	14.8x	0.3%	-3%	0%	-3%	
Samsung SDI	521,000	KRW	30.9	75.9x	15.7x	-1.5%	-4%	0%	148%	
CATL	293.5	CNY	206.3	14.2x	9.8x	-2.5%	-3%	-22%	-25%	
CALB	16.8	HKD	3.8	9.6x	2.4x	-4.6%	-12%	-13%	-47%	
EVE Energy	48.9	CNY	15.8	14.5x	9.2x	-2.2%	-5%	-10%	-40%	
Gotion	26.8	CNY	7.2	24.7x	5.9x	-1.9%	2%	-1%	-42%	
Sunwoda	20.1	CNY	5.5	18.5x	5.9x	2.9%	0%	13%	-40%	
Farasis	8.9	CNY	1.6	n.a.	n.a.	-1.6%	-2%	-7%	-56%	
Lithium Miners/Refiners										
Albemarle	108.3	USD	12.8	9.6x	4.6x	-4.6%	-2%	-19%	28%	
SQM	67.1	USD	18.5	9.4x	4.4x	-2.8%	-1%	-15%	51%	
Pilbara Minerals	3.9	AUD	8.7	21.2x	10.6x	-1.9%	-7%	-25%	61%	
Ganfeng Lithium	30.6	HKD	12.4	7.2x	6.7x	-2.5%	-10%	-25%	-20%	
Tianqi Lithium	40.9	CNY	9.9	10.1x	3.6x	-3.6%	-4%	-13%	-8%	
Cathode										
LG Chem	252,500	KRW	13.1	n.a.	2.4x	-0.6%	-5%	-5%	-14%	
EcoPro BM	105,000	KRW	7.7	531.9x	50.7x	0.7%	0%	-5%	-10%	
L&F	109,100	KRW	3.3	n.a.	15.2x	0.6%	-2%	-11%	42%	
Umicore	21.0	EUR	5.9	11.4x	4.9x	-2.0%	-3%	-10%	46%	
Ningbo Ronbay	23.4	CNY	2.5	41.7x	11.2x	-2.1%	-3%	-11%	-18%	
Beijing Easpring	37.4	CNY	3.0	18.2x	11.7x	-2.1%	-4%	-7%	-41%	
GEM	6.1	CNY	4.7	16.4x	5.1x	-1.6%	-2%	-6%	-22%	
Huayou Cobalt	35.4	CNY	10.0	8.8x	3.8x	-2.1%	-1%	-9%	-36%	
Dynanonic	40.1	CNY	1.7	13.4x	5.6x	-2.4%	-5%	-12%	3%	
Anode										
Posco	179,900	KRW	11.8	479.4x	47.4x	-2.3%	-2%	7%	23%	
Putailai	21.5	CNY	6.9	14.2x	8.8x	-2.5%	-3%	-9%	-23%	
Shanshan	11.5	CNY	3.9	n.a.	n.a.	-3.2%	3%	-4%	-19%	
Separator										
Asahi Kasei	1,649	JPY	14.2	15.2x	5.6x	2.6%	3%	1%	37%	
SK IE	17,170	KRW	1.0	n.a.	-21.0x	6.2%	9%	11%	-40%	
Toray Industries	1,256	JPY	11.9	23.4x	7.3x	0.4%	1%	-1%	30%	
W-Scope	152	JPY	0.1	n.a.	n.a.	-1.9%	1%	-1%	-36%	
Yunnan Energy	46.3	CNY	6.8	20.4x	7.9x	-2.5%	-5%	-14%	2%	
Shenzhen Senior	17.5	CNY	3.7	36.1x	11.4x	2.9%	10%	22%	27%	
Sinoma Science	57.2	CNY	14.3	31.8x	13.9x	-5.8%	-5%	17%	63%	
Electrolyte										
Enchem	20,200	KRW	0.3	n.a.	n.a.	0.0%	-1%	8%	-73%	
Tinci Minerals	31.8	CNY	9.6	10.3x	7.3x	-3.3%	-5%	-13%	-3%	
Capchem	70.6	CNY	7.9	23.7x	17.6x	-2.5%	-3%	16%	47%	
Jiangsu Guotai	7.5	CNY	1.8	9.5x	n.a.	-0.3%	-1%	-4%	-15%	
Shanshan	11.5	CNY	3.9	n.a.	n.a.	-3.2%	3%	-4%	-19%	
Equipment Maker										
Hirano	1,830	JPY	0.2	17.3x	n.a.	0.1%	0%	1%	11%	
CKD	5,240	JPY	2.2	28.2x	14.6x	0.8%	3%	-6%	87%	
Wuxi Lead	32.1	CNY	7.8	21.7x	17.3x	-0.9%	-1%	-4%	-51%	
Shenzhen Yinghe	20.9	CNY	2.0	20.3x	12.2x	-1.8%	-6%	0%	-40%	
Zhejiang HangKe	20.7	CNY	1.9	23.5x	19.2x	-2.3%	-1%	-7%	-46%	
Guangdong Lyric	37.8	CNY	0.9	40.8x	16.6x	-1.8%	5%	8%	-47%	
Shenzhen United	21.1	CNY	1.1	21.4x	13.1x	-2.4%	-1%	5%	-22%	

Source: Bloomberg (all estimates), Bernstein analysis
GLOBAL ENERGY STORAGE

Company Models

[CATL](#) | [LG Energy Solution](#) | [Samsung SDI](#)

[LG Chem](#) | [EcoPro BM](#) | [Posco Future M](#)

[Tianqi Lithium](#)

[Sungrow](#)

Industry Models

[Model: ESS tracker](#)

[Model: Monthly EV Tracker](#)

[Model: Battery Gigafactory Database](#)

[Industry model: Global EV TAM](#)

[Industry model: Global ESS TAM](#)

[Industry model: Battery TAM](#)

[Industry model: Battery supply and demand](#)

Blackbook

[Battery Utopia](#)

[Battery Tech Roadmap 2023: Faster, Higher, Stronger](#)

[Turn Up the Voltage... Investing Across the Battery Value Chain](#)

[Global Energy Storage: Resistance is Futile](#)

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[Future of Tech: Can China achieve AI supremacy?](#)

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[CATL: Initial perspectives on HK IPO Plans](#)

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[Global Energy Storage: Gigafactory expansion still strong but investment slowing](#)

[Global Energy Storage: Bigger and better batteries enables the rise of the 700km electric vehicle](#)

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BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	24 Sep 2026		TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
			Closing Price	Price Target		Cur	2025A	2026E	2027E	2025A	2026E	2027E
300750.CH (CATL)	O	CNY	293.50	800.00	(51.5)%	CNY	16.14	21.95	28.77	18.2	13.4	10.2
3750.HK (CATL)	O	HKD	492.80	770.00	(33.6)%	CNY	16.14	21.95	28.77	26.1	19.2	14.7
051910.KS (LG Chem)	M	KRW	252,500	312,000	(40.7)%	KRW (13,258.70)	2,693.22	28,743	(19.0)	93.8	8.8	
373220.KS (LGES)	M	KRW	351,500	351,000	(29.8)%	KRW (4,580.94)	(2,364.33)	10,225	(76.7)	(148.7)	34.4	
006400.KS (SDI)	M	KRW	521,000	520,000	120.4%	KRW (9,933.80)	1,751.66	18,577	(52.4)	297.4	28.0	
247540.KS (EcoPro BM)	U	KRW	105,000	140,000	(37.5)%	KRW	403.00	854.00	1,963.00	260.5	123.0	53.5
300274.CH (Sungrow)	O	RMB	84.21	151.00	(74.4)%	RMB	6.55	6.93	7.95	12.9	12.2	10.6
002466.CH (Tianqi Lithium)	O	CNY	40.90	80.00	(35.5)%	CNY	0.28	3.90	7.75	144.9	10.5	5.3
9696.HK (Tianqi Lithium)	O	HKD	29.20	65.00	(54.1)%	CNY	0.28	3.90	7.75	88.6	6.4	3.2
ASIA			1,978.05									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

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VALUATION METHODOLOGY

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